MIL–Q–12713C
23 NOVEMBER 1960
SUPERSEDING
MIL–Q–12713B(Ord)
8 DECEMBER 1959

MILITARY SPECIFICATION

# QUADRANT, FIRE CONTROL: GUNNER'S M1A1

*This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.*

## 1. SCOPE

**1.1** This specification covers one type of quadrant, known as Gunner's Fire Control Quadrant, M1A1, used primarily for measuring angles of elevation for various high and low trajectory weapons.

## 2. APPLICABLE DOCUMENTS

**2.1** The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

### SPECIFICATIONS

MILITARY

MIL–F–13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL–F–14619 —Fire Control Parts Equipment and Tools; Packaging of.

### STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–130 —Identification Marking of U.S. Military Property.

### DRAWINGS

ORDNANCE CORPS

F8228867 —Quadrant, Fire Control: Gunner's, M1A1

LSQAP–8228867—List of Supplementary Quality Assurance Provisions.

IEL 8228867 —Index of Inspection Equipment Lists, Quadrant, Fire Control: Gunner's, M1A1.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Production model.** At the beginning of production one quadrant, representative of production, shall be made available to the Government to determine the ability of the quadrant to meet all requirements of this specification.

**3.2 Materials.** Materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

**3.3 Construction.** The quadrant shall be manufactured in accordance with Drawing F8228867 "Quandrant, Fire Control: Gunner's M1A1" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance, and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

FSC 1290

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**3.4 General specification.** The quadrant shall meet the following requirements of Specification MIL–F–13926.

(a) Order of precedence.
(b) Protective finished surface.
(c) Special accuracy.
(d) Workmanship.

**3.5 Environmental.**

**3.5.1** *Operating temperatures.* The quadrant shall meet the requirements as specified in 3.7 while exposed to, and thermally stabilized at ambient temperatures of 150 and minus 40° Fahrenheit (F).

**3.6 Performance.**

**3.6.1** *Shoe contact and location.* Each set of shoes shall show full contact over their bearing surface when tested on a flat surface plate. The bearing surfaces of the two sets of shoes shall be perpendicular to each other within the tolerance specified on Drawing C7672992.

**3.6.2** *Index alinement.* With the index pointer on the 0–800 mil side set to coincide with the "0" graduation on that scale, and the micrometer reading of zero, the following conditions shall be met:

(a) The "0" index line on the arm shall be in coincidence with the index line on the plunger.
(b) The index pointer on the 800–1600 mil side shall coincide with the 1,500-mil graduation.
(c) The micrometer shall indicate number 10.

**3.6.3** *Level bubble.* The level vial shall be adjusted so that the bubble will be centered with respect to the graduations on the vial within a tolerance of plus or minus 0.4 mil (1 minute 21 seconds) when:

(a) The locating shoes, for use with the 0–800-mil scale, are in a true horizontal plane and:
  (1) The index pointer for the 0–800 mil scale coincides with the 0 graduation and the micrometer reading is zero.
  (2) The index line on the arm and plunger are in coincidence on the 0–800 mil side.
(b) The locating shoes, for use with 800–1600-mil scale, are in a true 45° plane and:
  (1) The index pointer for the 800–1600-mil scale coincides with the 800 graduation and the micrometer reading is zero.
  (2) The index line on the arm and plunger are in coincidence on the 800–1600-mil side.

**3.6.4** *Accuracy.* In any setting of the quadrant, the error shall not exceed 0.4 mil.

**3.7 Torque.** Torque on the micrometer knob shall not exceed 1 pound-inch at normal room temperature and shall not exceed 2 pound-inches at the extreme operating temperatures as specified in 3.5.1.

**3.8 Identification marking.** Unless otherwise specified, identification marking shall be in accordance with Standard MIL–STD–130.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Contractor inspection.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.2 Contractor's quality assurance system.** The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the



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applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

(e) Measures for preventing any significant departure of product quality from standards.

**4.2.1** The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure required by this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

**4.3 Government verification.** All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have a detrimental affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

**4.4 Production model.** Unless otherwise specified in the contract, one quadrant shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

**4.4.1** *Production model failure.* Should the production model fail to pass the prescribed examinations and tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

**4.5 Inspection provisions.**

**4.5.1** *Lot formation.* Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL–STD–105.

**4.5.2** *Inspection of components.* Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP)—8228867.

**4.5.2.1** *Contents of SQAP.* The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications, and to reject individual non-conforming items.

**4.5.2.2** *Contractor's inspection for all components not covered by SQAP.* Inspection for these components shall be in accordance with the inspection procedures of the written inspection plan required by 4.2 and 4.2.1. The plan shall include procedures for checking all positional and functional characteristics shown on the drawings



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which indicate precision requirements and have a bearing on the functional accuracy and performance of the quadrant.

**4.5.2.3** *Disposition of nonconforming product.* Disposition of nonconforming components shall be in accordance with Standard MIL–STD–105.

**4.6 Control of materials.** Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

**4.7 Examination.** The classification of defects in tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance.

**4.7.1** All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government informed of the corrective action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL–STD–105.

**4.8 Inspection equipment.**

**4.8.1** *Contractor provisions.* Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

**4.8.2** *Government provisions.* Test equipment listed in IEL–8228867, when available, will be

TABLE I. *Classification of defects*

Use inspection level II, sampling plan table IV–A of Standard MIL–STD–105

*Critical:* Nondefined.

| *Major: AQL 1.0%* | | *Requirement paragraph* | *Q.A. paragraph* | *Test method* |
|---|---|---|---|---|
| 101 | Completeness of assy | 3.3 | ..... | Visual |
| 102 | Workmanship | 3.4 | ..... | MIL–F–13926 |
| 103 | Shoe contact and location | 3.6.1 | 4.9.1 | Standard measuring Equipment (SME) |
| 104 | Index alignment | 3.6.2 | 4.9.2 | Tactile-visual |
| 105 | Level vial adjustment | 3.6.3(a) and (b) | 4.9.3 | Device, special testing (DST) |
| 106 | Accuracy | 3.6.4 | 4.9.3.4 | DST |
| 107 | Torque | 3.7 | 4.9.4 | SME |
| 108 | Identification marking | 3.8 | 4.9.5 | Visual |

*Minor:* None defined.

TABLE II. *Classification of defects*

Use inspection level I, and sampling plan Table V of Standard MIL–STD–105

*Critical:* Nondefined.

| *Major: AQL 1.5%* | | *Requirement paragraph* | *Q.A. paragraph* | *Test method* |
|---|---|---|---|---|
| 101 | Temperature | 3.5.1 | 4.9.6 | SME |
| 102 | Torque | 3.7 | 4.9.4 | SME |

*Minor:* None defined.



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used for Government product inspection (reference 4.3), and to settle disputes (reference 4.2.1). Drawings of this equipment will be furnished the contractor for informational purposes.

**4.8.3** *Accuracy requirments.* Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance except in those cases specified otherwise. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

**4.8.4** *Accuracy requirements for special testing device.*

**4.8.4.1** *Device to test temperature requirements.* The temperature test chamber shall meet the quality assurance requirements for volume, measurments and tolerance specified in Specification MIL–F–13926.

**4.8.4.2** *Device to test accuracy and level vial requirements.* The device for testing accuracy and level vial requirements shall be equipped with suitable means to permit mounting the quadrant as specified. Provisions shall be made on the device to permit reading errors of the quadrant at any settings on the 0–800-mil scale and 800–1600 mil scale, the cumulative error of which shall not exceed 0.25 mil throughout the entire excursion. Rigidity, stability, ease of maintenance, and adaptability of setup or calibration equipment, with complete provisions, are also prime requirements. All special setup or calibration equipment necessary for certification of device accuracy shall be furnished by the contractor and will become part of the device.

**4.9 Test methods and procedures.**

**4.9.1** *Shoe contact and location.* The contact area and perpendicularity of the two sets of shoes shall be inspected on a flat surface plate by means of standard measuring equipment to determine compliance with the requirements specified.

**4.9.2** *Index alinement.* Index alinement shall be visually inspected at the specified settings of the index pointer micrometer and arm and plunger, to determine compliance with the requirements specified.

**4.9.3** *Government verification and Government product inspection.* Government verification testing to determine proper level vial adjustment and accuracy shall be performed utilizing the test equipment specified on the IEL listed in section II.

**4.9.3.1** This test equipment shall be adjusted as outlined in the setup and operating procedures that accompany the test equipment.

**4.9.3.2** Set the quadrant under test to zero mils elevation and position the quadrant on the adapter and observe the level vial setting. Both level vials shall show identical settings. Any level vial setting of the quadrant under test not in coincidence with the calibrated quadrant, shall be adjusted to indicate identical level positions by rotating the micrometer of the calibrated quadrant or by rotating the test fixture (by means of the handwheel) and the calibrated quadrant's micrometer until coincidence is obtained. The amount of error shall be read from the calibrated quadrant plus or minus the correction factor. This error shall not exceed the limits specified in 3.6.3.

**4.9.3.3** Adjust the fixture as specified in the setup and operating procedure and proceed with the test as outlined in 4.9.3.2; however, the 800-mil graduation will be used in lieu of zero elevation. This error shall not exceed the limits specified in 3.6.3.

**4.9.3.4** Accuracy shall be tested by utilizing the setup and test procedure outlined in 4.9.3.2. In addition, the inspector shall select four other angles of elevation for each scale. These additional settings shall change at the discretion of the inspector.

**4.9.4** *Torque.* This test will be performed utilizing a standard torque wrench and appropriate adapter. When performing this test, the



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initial reading obtained to actuate the micrometer shall be disregarded.

**4.9.5** *Identification marking.* Identification marking shall be visually inspected for conformance with Standard MIL–STD–130.

**4.9.6** *Temperature test.* The quadrant shall be subjected to the test procedure outlined under procedure I of Specification MIL–F–13926, utilizing the temperature specified in 3.5. At the high and low temperature, the quadrant shall conform to the requirement of 3.7 when tested as outlined in 4.9.4. Samples selected for this test shall be selected from the samples that have been subjected to the acceptance test outlined herein.

**4.9.7** *Packaging, packing, and marking.* Quality assurance provisions of Specification MIL–F–14619 shall apply.

## 5. PREPARATION FOR DELIVERY

**5.1 Packaging, packing, and marking.** Packaging, packing, and marking shall be in accordance with Specification MIL–F–14619/P12728. The level of protection shall be as specified in the procurement document.

## 6. NOTES

**6.1 Intended use.** The quadrant is intended to lay various guns and field pieces (towed or self-propelled) in elevation and to measure elevation angles.

**6.2 Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

**6.3 Design information.** Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

**6.4 Definitions.**

**6.4.1** *Standard measuring equipment (SME).* Standard measuring equipment includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

**6.4.2** *Device, special testing (DST).* Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

**6.4.3.** *Tactile.* Skillful manipulation or sensing.

**Notice.** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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Army—Ordnance Corps
Navy—Marine Corps
Air Force—WRA

**Preparing activity:**
Army—Ordnance Corps

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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